

Subspaces continued...

Recall:

Definition 31. A subset S of a vector space V (over F) is called a subspace if

- S is non-empty;
- S is closed under addition;
- S is closed under scalar multiplication.

Theorem 12. A subspace S of vector space V is a vector space in its own right.

Proof. Since we have that S is non-empty, there exists $u \in S$, but S is also closed under scalar multiplication and so $0u = \mathbf{0} \in S$, we also have that for each $u \in S$, $(-1)u = -u \in S$. The other vector space axiom identities hold in particular for S since they hold for all elements in V . $\square$

More examples: (3) Let $V = \mathbb{R}^n$ with component-wise addition and scalar multiplication. $\{\mathbf{0}\}$ and $V = \mathbb{R}^n$ itself are always subspaces.

For $\mathbf{0} \neq \mathbf{v} \in \mathbb{R}^n$, let $L_{\mathbf{v}} = \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{x} = \lambda \mathbf{v} \text{ for some } \lambda \in \mathbb{R}\}$, notice that $\mathbf{0} \in L_{\mathbf{v}}$, and if $\mathbf{x}, \mathbf{y} \in L_{\mathbf{v}}$, we have that $\mathbf{x} = \lambda_1 \mathbf{v}$ and $\mathbf{y} = \lambda_2 \mathbf{v}$ and so $\mathbf{x} + \mathbf{y} = (\lambda_1 + \lambda_2)\mathbf{v}$, thus $\mathbf{x} + \mathbf{y} \in L_{\mathbf{v}}$. Similarly for scalar $\mu \in \mathbb{R}$, we have $\mu \mathbf{x} = \mu(\lambda_1 \mathbf{v}) = (\mu \lambda_1)\mathbf{v}$ and so $\mu \mathbf{x} \in L_{\mathbf{v}}$, that is $L_{\mathbf{v}}$ is a subspace of $\mathbb{R}^n$ (with component-wise addition and scalar multiplication).

$P = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n \mid a_1 x_1 + a_2 x_2 + \dots + a_n x_n = 0 \text{ where } a_1, a_2, \dots, a_n \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^n$ (with component-wise addition and scalar multiplication). Check this.

Taking $n = 3$, the above tells us that lines and planes through the origin are subspaces of $\mathbb{R}^3$ (with component-wise addition and scalar multiplication).

Remark 2. For $1 \leq k < n$, $\mathbb{R}^k$ is not a subspace of $\mathbb{R}^n$, note that it is not even a subset - the former is a set of k -tuples while the latter a set of n -tuples, however $\mathbb{R}^k$ can be 'identified' with a subspace of $\mathbb{R}^n$ given by $S = \{(x_1, x_2, \dots, x_n) \in \mathbb{R}^n \mid x_n = x_{n-1} = \dots = x_{k+1} = 0\}$, that is a set of all n -tuples whose last $n - k$ components are 0.

(4) Let $D \subseteq \mathbb{R}$ and consider $V = \mathbb{R}^D$, the set of all real valued functions $f : D \rightarrow \mathbb{R}$ with domain D with addition and scalar multiplication defined input-wise, that is for $f, g \in \mathbb{R}^D$ and $\lambda \in F$, $(f + g)(x) = f(x) + g(x)$ and $(\lambda f)(x) = \lambda f(x)$ for each $x \in D$.

Consider $D \in \mathbb{R}$ such that $\pi \in D$, let $S = \{f \in \mathbb{R}^D \mid f(\pi) = 0\}$, notice that the 0 constant function is in S since it will in particular maps π to 0. Now suppose that $f, g \in S$, and then we have that $(f + g)(\pi) = f(\pi) + g(\pi) = 0 + 0 = 0$ and so $f + g \in S$, similarly for scalar $\lambda \in \mathbb{R}$ we have $(\lambda f)(\pi) = \lambda f(\pi) = \lambda \cdot 0 = 0$.

Take $D = \mathbb{R}$, and let $T = \{f \in \mathbb{R}^{\mathbb{R}} \mid f \text{ is differentiable and } \frac{d}{dx}f + 4f = 0\}$, Notice that the 0 constant function is in T and for $f, g \in T$ we have that $f + g$ is also differentiable and $\frac{d}{dx}(f + g) + 4(f + g) = \frac{d}{dx}f + \frac{d}{dx}g + 4f + 4g = (\frac{d}{dx}f + 4f) + (\frac{d}{dx}g + 4g) = 0 + 0 = 0$, check similarly that T is closed under scalar multiplication.

Side note: In some sense the examples considered are not far from each other - $M_{m \times n}(\mathbb{R})$ can be ‘identified’ with $\mathbb{R}^{n^2}$, and $\mathbb{R}^n$ ‘identified’ with $\mathbb{R}^D$ with $D = \{1, 2, \dots, n\}$.

Non-examples (1) Consider $\mathbb{R}^3$ with component-wise addition and scalar multiplication. The set $N = \{(x_1, x_2, x_3) \in \mathbb{R}^3 \mid 2x_1 - 100x_2 + 5x_3 = 4\}$ is not a subspace. Notice that $(0, 0, 0) \notin N$, for if it were the case we would have $0 = 4$.

(2) Consider $M_{n \times n}(\mathbb{R})$ with component-wise addition and scalar multiplication.

The set $M = \{A \in M_{n \times n}(\mathbb{R}) \mid \det(A) = 0\}$ is not a subspace. Notice that even though we have the zero matrix $\mathbf{0} \in M$ and that for $\lambda \in \mathbb{R}$ and $A \in M$ we have $\lambda A \in M$ since $\det(\lambda A) = \lambda^n \det(A) = \lambda^n \cdot 0 = 0$, M is not closed under addition. Take A and B with coefficients

$$\text{given by } a_{ij} = \begin{cases} 1 & \text{if } i = j \text{ and } 1 \leq i \leq n-1 \\ 0 & \text{otherwise.} \end{cases} \text{ and } b_{ij} = \begin{cases} 1 & \text{if } i = n = j \\ 0 & \text{otherwise.} \end{cases}$$

Matrices A and B are diagonal matrices with at least one zero on the diagonal and so $\det(A) = 0 = \det(B)$, however $A + B = I_n$ and so $\det(A + B) = \det(I_n) = 1 \neq 0$.

Is a union two of subspaces always a subspace?

That is, if S_1 and S_2 are subspaces of vector space V (over F), is $S_1 \cup S_2$ a subspace of V ? No.

Consider $\mathbb{R}^2$ with component-wise addition and scalar multiplication, and take subspaces

$S_1 = \{(x, y) \mid x = 0\}$ and $S_2 = \{(x, y) \mid y = 0\}$, notice that $(0, 3150), (3150, 0) \in S_1 \cup S_2$, however $(0, 3150) + (3150, 0) = (3150, 3150) \notin S_1 \cup S_2$.

Proposition 14. Let I be a set and S_i a subspace of vector space V (over F) for each $i \in I$, then intersection $\cap_{i \in I} S_i$ is a subspace of V .

Proof. Note that $0 \in \cap_{i \in I} S_i$ since $0 \in S_i$ (for each $i \in I$). Now take $x, y \in \cap_{i \in I} S_i$, we then have that $x, y \in S_i$ for each $i \in I$, but since each S_i is a subspace of V we have that $x + y \in S_i$ for each $i \in I$ and thus $x + y \in \cap_{i \in I} S_i$. Similarly, let $\lambda \in F$ and $x \in \cap_{i \in I} S_i$, then $x \in S_i$ for each $i \in I$ and so $\lambda x \in S_i$ for each $i \in I$, that is, $\lambda x \in \cap_{i \in I} S_i$. $\square$

In particular if we take $I = \{1, 2\}$, S_1 and S_2 subspaces of vector space V , then their intersection $S_1 \cap S_2$ is also a subspace of V (if the above proof is not clear, try proving it for two subspaces S_1 and S_2).

Extra note: The last proposition tells us that arbitrary intersections of subspaces is again a subspace. This suggest for any subset of our vector space, there exists a smallest subspace containing that set (take intersection of all subspaces containing that set) and this smallest subspace turns out to have a simple discription in terms of ‘linear combinations’ which we shall study in next lectures.